



**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
*Off-Campus Centre of Nitte (Deemed to be University)*  
**Fourth Semester B.Tech (CBCS) Degree Examinations**

May 2024

**EC2102-1 – ELECTROMAGNETIC WAVE THEORY**  
 (E&C)

Duration: 3 Hours

Max. Marks: 100

**Note:**

Part – A: Multiple Choice Questions: Answer all Twenty questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

**PART - A: MULTIPLE CHOICE QUESTIONS**

**20 Marks**

- The continuity equation is a combination of which of the following.
 

|                             |                               |
|-----------------------------|-------------------------------|
| A) Ohm's law and Gauss law  | B) Ampere law and Gauss law   |
| C) Ohm's law and Ampere law | D) Maxwell law and Ampere law |
- The potential at a point (4, 3, -6) for the function  $V = 2x^2y + 5z$  volts is?
 

|        |         |
|--------|---------|
| A) 96V | B) 36V  |
| C) 66V | D) -66V |
- The force on a charge of 2C in a field 1V/m is \_\_\_\_\_
 

|       |       |
|-------|-------|
| A) 0N | B) 1N |
| C) 2N | D) 3N |
- The Gaussian surface for a point charge will be
 

|           |             |
|-----------|-------------|
| A) Cube   | B) Cylinder |
| C) Sphere | D) Cuboid   |
- Dielectric constant will be high in
 

|               |                    |
|---------------|--------------------|
| A) Conductors | B) Semiconductors  |
| C) Insulators | D) Superconductors |
- The work done by a charge of  $10\mu\text{C}$  with a potential 4.386 is (in  $\mu\text{J}$ )
 

|                        |                        |
|------------------------|------------------------|
| A) 32.86 $\mu\text{J}$ | B) 43.86 $\mu\text{J}$ |
| C) 54.68 $\mu\text{J}$ | D) 65.68 $\mu\text{J}$ |
- The permittivity is also called
 

|                         |                        |
|-------------------------|------------------------|
| A) Electrostatic energy | B) Dielectric constant |
| C) Dipole moment        | D) Susceptibility      |
- Gauss law cannot be used to find which of the following quantity.
 

|                             |                          |
|-----------------------------|--------------------------|
| A) Electric field intensity | B) Electric flux density |
| C) Charge                   | D) Permittivity          |
- Calculate the magnetic field at a point on the centre of the circular conductor of radius 0.5m with current 1.422A.
 

|                 |                 |
|-----------------|-----------------|
| A) 0.01A/m unit | B) 0.452 A/m    |
| C) 3 A/m        | D) 4.6 A/m unit |
- The total permeability in a conductor is
 

|                                                  |                          |
|--------------------------------------------------|--------------------------|
| A) Absolute permeability                         | B) Relative permeability |
| C) Product of absolute and relative permeability | D) Unity                 |
- When a dielectric loses its dielectric property, the phenomenon is called
 

|                    |                         |
|--------------------|-------------------------|
| A) Dielectric Loss | B) Dielectric Breakdown |
| C) Polarization    | D) Magnetization        |
- The total polarisation of a material is the
 

|                                          |                                         |
|------------------------------------------|-----------------------------------------|
| A) Product of all types of polarisation  | B) Sum of all types of polarisation     |
| C) Orientation directions of the dipoles | D) Total dipole moments in the material |

13. The wavelength of a wave in a waveguide  
 A) is greater than of free space  
 B) depends only on the waveguide dimensions and the free-space wavelength  
 C) is inversely proportional to the phase velocity  
 D) is directly proportional to the group velocity
14. An example for electromagnetic wave propagation is  
 A) refrigerator  
 B) electric fan  
 C) mobile transponder  
 D) relays in actuators
15. The expression for velocity of a wave in the conductor is  
 A)  $V = \sqrt{2\omega/\mu\sigma}$   
 B)  $V = \sqrt{(2\omega\mu\sigma)}$   
 C)  $V = (2\omega/\mu\sigma)$   
 D)  $V = (2\omega\mu\sigma)$
16. The Smith chart is a polar chart which plots  
 A) R vs Z  
 B) R vs  $Z_{norm}$   
 C) T vs Z  
 D) T vs  $Z_{norm}$
17. In a shorted line, the reflection coefficient will be  
 A) 0  
 B) 1  
 C) -1  
 D) Infinity
18. The range of the standing wave ratio is  
 A)  $0 < S < 1$   
 B)  $1 < S < 1$   
 C)  $1 < S < \infty$   
 D)  $0 < S < \infty$
19. The propagation constant of a wave with attenuation and phase constants are 2 and 3 respectively is  
 A)  $2 - 3j$   
 B)  $3 - 2j$   
 C)  $2 + 3j$   
 D)  $3 + 2j$
20. Moving towards the clockwise direction in the Smith chart implies moving  
 A) Towards generator  
 B) Towards load  
 C) Towards stub  
 D) Towards waveguide

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I |                                                                                                                                                                                                                                                                                              | Marks | BT* | CO* | PO* |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.       | a) A total charge of 50nC is uniformly distributed around a ring of radius 3m with its center located at origin and is in the Z=0 plane. Determine the electric potential at (0,0,4). Also determine the potential at (0,0,4), if all charges of 50nC were to be concentrated at the origin. | 6     | L*3 | 1   | 1   |
|          | b) Explain and derive boundary conditions for the interface between conductor and free space with a neat diagram.                                                                                                                                                                            | 6     | L2  | 1   | 1   |
|          | c) Define electric field intensity. Obtain an expression for electric field intensity at a point due to many point charges.                                                                                                                                                                  | 4     | L2  | 1   | 1   |
| 2.       | a) State Gauss law and prove Divergence theorem.                                                                                                                                                                                                                                             | 8     | L2  | 1   | 1   |
|          | b) The charges $Q_1 = -1\mu\text{C}$ , $Q_2 = 4\mu\text{C}$ and $Q_3 = 3\mu\text{C}$ are in free space at (0, 0, 0), (0, 0, 1) and (1, 0, 0) respectively. Find the total energy stored in the system.                                                                                       | 8     | L3  | 1   | 1   |
| 3.       | a) State and Derive Amperes circuital law in integral form.                                                                                                                                                                                                                                  | 4     | L2  | 2   | 1   |
|          | b) Determine magnetic field Intensity at a point using Biot Savart's law due to a current in a straight conductor of finite length.                                                                                                                                                          | 7     | L2  | 2   | 1   |
|          | c) Magnetic field Intensity in free space is given by:<br>$H = 10\rho^2 (\vec{a}\Phi)^\wedge \text{A/m,}$ i) Find J A/m <sup>2</sup><br>ii) Obtain total current by integrating J over a circular surface $\rho = 1$ , $Z = 0$ and all $\phi$ .                                              | 5     | L3  | 2   | 1   |

## Unit – II

|    |     |                                                                                                                                                                                                      |   |    |   |   |
|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| 4. | a)  | State and prove Pointing's theorem for electromagnetic field.                                                                                                                                        | 7 | L2 | 4 | 1 |
|    | b)  | A plane wave travelling in positive 'x' direction in a lossless unbounded medium having permeability 4.5 times that of free space and a permittivity twice that of the wave.                         |   |    |   |   |
|    | i)  | Find phase velocity of the wave.                                                                                                                                                                     |   |    |   |   |
|    | ii) | Find the amplitude and direction of 'H', if 'E' has only 'y' component with amplitude 20 V/m.                                                                                                        | 5 | L3 | 4 | 1 |
|    | c)  | A circular conducting loop of wire of 10cm radius is located on XY plane with magnetic flux density $B = 0.5 \cos(377t) (3\hat{a}_y + 4\hat{a}_z)$ Tesla. Determine the voltage induced by the path. | 4 | L3 | 3 | 1 |
| 5. | a)  | Find the amplitude of displacement current density $J_D$ A/m <sup>2</sup>                                                                                                                            |   |    |   |   |
|    | i)  | In air $E = 80 \cos(6.277 \times 10^8 t - 2.092y) \hat{a}_z$ V/m                                                                                                                                     |   |    |   |   |
|    | ii) | $D = 3 \times 10^{-6} \sin(6 \times 10^6 t - 0.3464x) \hat{a}_z$ C/m <sup>2</sup> with relative permeability of 600.                                                                                 | 6 | L4 | 3 | 1 |
|    | b)  | Do the following fields satisfy Maxwell's equations?                                                                                                                                                 |   |    |   |   |
|    | i)  | $E = E_m \sin(x) \sin(t) \hat{a}_y$ V/m                                                                                                                                                              | 6 | L3 | 3 | 1 |
|    | ii) | $H = (E_m / \mu_0) \cos(x) \cos(t) \hat{a}_z$ A/m.                                                                                                                                                   | 4 | L2 | 3 | 1 |
|    | c)  | Derive Maxwell's equation for time varying fields from Faraday-Lenz Law                                                                                                                              |   |    |   |   |
| 6. | a)  | Derive an expression for wave propagation in free space.                                                                                                                                             | 6 | L2 | 4 | 1 |
|    | b)  | Describe wave polarization. Derive an expression for wave polarization.                                                                                                                              | 6 | L2 | 4 | 1 |
|    | c)  | Define depth of penetration. For silver, the conductivity is $3.0 \times 10^6$ U/m. At what frequency will the depth of penetration be 1mm.                                                          | 4 | L3 | 4 | 1 |

## Unit – III

|    |      |                                                                                                                                                               |   |    |   |   |
|----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| 7. | a)   | Derive the general transmission line equation for voltage at any point on a transmission line.                                                                | 8 | L2 | 5 | 1 |
|    | b)   | Derive the expression for input impedance for a terminated loss less transmission line and derive the same for quarter wave line.                             | 8 | L2 | 5 | 1 |
| 8. | a)   | Explain the construction of Smith chart for constant resistance circle with a neat diagram and relevant expressions.                                          | 8 | L2 | 5 | 1 |
|    | b)   | The $0.1 \lambda$ length line shown has a characteristic impedance of $50 \Omega$ and is terminated with a load impedance of $Z_L = 5 + j25 \Omega$ . What is |   |    |   |   |
|    | i)   | The impedance at $L = 0.1 \lambda$ ,                                                                                                                          |   |    |   |   |
|    | ii)  | VSWR on the line and                                                                                                                                          |   |    |   |   |
|    | iii) | Reflection coefficient from the load.                                                                                                                         | 8 | L4 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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